ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

IMPRESSIVE PERSONALITY



James Dyson is a British inventor, engineer, entrepreneur, industrial designer and philanthropist who is widely recognised for challenging conventional engineering solutions and transforming practical problems into innovative products. His career is particularly inspiring because it brings together creativity, design, engineering, experimentation, business and education. Born in Norfolk, England, in 1947, Dyson developed an early interest in activities such as sailing, model making and long-distance running. His father, Alec Dyson, was a teacher at Gresham's School, where James studied before pursuing art and design. He attended the Byam Shaw School of Art and later the Royal College of Art, where he studied architectural design. Although his educational journey did not begin through a traditional engineering route, his exposure to art and design helped him develop a unique approach to problem-solving by combining creativity with practical functionality. During his time at the Royal College of Art, Dyson became interested in functional product design and was introduced to inventor and engineer Jeremy Fry. Fry later invited him to work on the Sea Truck, a high-speed landing craft, which provided Dyson with valuable practical experience in engineering, manufacturing, materials and commercial product development. This experience helped him understand that successful engineering requires more than simply having an innovative idea; a design must also be practical, reliable and effective in the real world.

Dyson later developed the Ballbarrow, an innovative alternative to the traditional wheelbarrow. Instead of accepting the conventional narrow wheel as the only solution, he replaced it with a large pneumatic ball that distributed the load more effectively and reduced the tendency of the wheelbarrow to sink into soft ground. This early invention demonstrated one of Dyson's most important characteristics: his willingness to question existing designs and search for better solutions. His most famous engineering journey began in 1978 when he became frustrated with the declining performance of his Hoover Junior vacuum cleaner as its dust bag became filled. Rather than accepting this as an unavoidable limitation, Dyson investigated the underlying problem and became interested in cyclone technology after observing a cyclone system at a sawmill. He began exploring whether a cyclone could separate dust from air and eliminate the need for a conventional vacuum bag. Dyson then began a

long process of experimentation and prototype development. Over approximately five years, he produced 5,127 prototypes before achieving a successful design. This remarkable process represents his belief that failure is not the end of innovation but an important part of learning and improvement. His engineering method can be represented as Problem → Idea → Prototype → Test → Failure or Feedback → Modification → Improvement. Instead of becoming discouraged when an experiment failed, Dyson used the results to understand what needed to be changed and continued developing the design.

His early cyclone technology was eventually commercialised in Japan as the G-Force vacuum cleaner, and in 1993 the Dyson DC01 was launched under his own name. The DC01 introduced cyclone technology through a bagless vacuum system and became a major commercial success. Dyson's achievement was therefore much more than creating another household appliance; he challenged an established design principle and introduced a different technological approach to solving a common problem. His success demonstrates that innovation often begins by questioning something that people have accepted for a long time and asking whether it can be redesigned in a better way. Following the success of his vacuum technology, Dyson expanded into several areas, including air treatment, hand dryers, hair-care technology, high-speed electric motors, robotics, batteries and other advanced technologies. His products reflect his focus on airflow, engineering efficiency, compact technology, usability and functional design. His work also demonstrates the importance of transferring knowledge between different engineering fields. For example, technologies and principles related to motors, airflow, sensors and computational systems can be applied to different products and challenges. His interest in robotics further demonstrates how modern engineering brings together mechanical systems, electronics, software, sensors, computer vision, artificial intelligence and machine learning.

Another important characteristic of Dyson is his ability to combine different disciplines. His background in art and design became valuable when combined with engineering, allowing him to consider not only how a product functions but also how people interact with it. This combination of design and engineering became an important feature of Dyson's products, where functionality, usability, appearance and technological performance are considered together. Dyson also demonstrated strong entrepreneurial thinking because he did not stop after developing an invention. He worked to transform his technology into commercially successful products and eventually built Dyson into a global technology company. His career therefore connects technical innovation with manufacturing, intellectual property, product development, business strategy and entrepreneurship. Beyond his commercial

achievements, Dyson has also contributed significantly to engineering education and research. The James Dyson Foundation was established to encourage young people to study science, technology, engineering and mathematics and supports educational activities and the James Dyson Award, which encourages students to develop innovative engineering solutions. Dyson also established the Dyson Institute of Engineering and Technology, which combines academic learning with practical industry experience. His association with Imperial College London has further supported areas such as design engineering, robotics and technological research. These efforts show that Dyson's interest in engineering extends beyond his own inventions, as he also wants to encourage future generations to become engineers, designers and innovators.

Dyson has received significant recognition for his contributions to engineering and innovation, including a CBE in 1996, a knighthood in 2007, election as a Fellow of the Royal Society in 2015 and appointment to the Order of Merit in 2016. However, the most impressive aspect of his career is not simply his awards or commercial success but the mindset behind his achievements. His personality can be characterised by curiosity, persistence, creativity, critical thinking, problem-solving, experimentation, willingness to challenge assumptions, attention to functionality, interdisciplinary thinking and entrepreneurship. He demonstrates that an engineer should not simply accept existing solutions but should continuously question whether there is a better way. His story of developing 5,127 vacuum prototypes is a powerful example of how persistence and continuous improvement can transform repeated failures into meaningful innovation. His journey is particularly relevant to me because I am interested in technology, engineering, hardware, AI, cybersecurity, IoT and understanding how systems work. Dyson's approach encourages me to look beyond simply learning how to use technology and instead understand the principles behind it, experiment with different solutions, build practical systems and continuously improve my skills. His career also shows me that technical knowledge becomes more powerful when combined with creativity, communication, design thinking and business awareness. The most important lesson I learn from Sir James Dyson is that innovation begins with curiosity but becomes meaningful through action. A good engineer must be willing to identify real-world problems, question conventional solutions, develop ideas, create prototypes, test them, learn from failures and continue improving until a better solution is achieved. For these reasons, I have chosen Sir James Dyson as my impressive personality because he represents the kind of engineer I aspire to become: someone who does not simply accept the way things are, but understands them, questions them, experiments with them and works persistently to make them better. His journey can ultimately be represented as

Observe → Question → Understand → Invent → Prototype → Test → Learn → Improve → Innovate, making Sir James Dyson an inspiring example of how engineering, creativity, persistence and entrepreneurship can come together to create meaningful technological change.

Self Introduction: Written Script and One-Page Profile

T. Harshavardhan

Aspiring Cybersecurity and AI/ML Specialist

Links	LinkedIn: https://www.linkedin.com/in/harsha-vardhan08/
Summary	I like solving problems and I don’t give up until they are fixed. My goal is to work in cybersecurity and AI/ML, building smart systems that keep people safe online. I’m starting with C, HTML and JavaScript.
Introduction	Hello, I’m T. Harshavardhan, a first-year Computer Science Engineering student with a clear target: a career in cybersecurity and AI/ML. I like knowing what is happening under the hood, from how memory is managed in C to how a browser talks to a server, because that is where vulnerabilities live. I’m currently planning a spam call detection project to protect ordinary users from phone fraud, which brings both of my interests together. I’m ambitious about where this leads: first a skilled security engineer, then a specialist in AI-driven security that others trust to keep their systems safe.
Skills	C, HTML, JavaScript, Web Development
Education	B.E. Computer Science and Engineering, 1st Year

SWOC

SELF ANALYSIS

EN25C09

MY PERSONAL PROFILE

A creative reflection on my skills, interests, growth areas and future goals

Strengths • Weaknesses • Opportunities • Challenges

I like combining creativity with technology.

01 INTRODUCTION

A SWOC analysis is a simple way to understand myself from four perspectives: my strengths, the areas I need to improve, the opportunities available to me, and the challenges I need to overcome. This analysis helps me identify my present abilities and create a clearer direction for my personal, academic and professional growth.

STRENGTHS

- Quick learner
- Good at teamwork
- Patient
- Curious to learn
- Creative

WEAKNESSES

- Stage fear
- Overthinking
- Getting distracted
- Public speaking

OPPORTUNITIES

- Learning programming languages like C
- Improving communication and English skills
- Learning through YouTube and other online resources
- Participating in hackathons and ideathons
- Learning from teachers and seniors

CHALLENGES

- Managing time between academics and skill development
- Lack of practical experience
- Maintaining focus
- Finding the right career path
- Completing projects on time
- Communicating with new people

02 MY INTERESTS & DIRECTION

- **Programming & Coding**
- **Creating New**
- **Team Projects**
- **Business & Entrepreneurship**
- **Technology & AI**
- **Problem Solving**
- **Designing & Creativity**
- **Cybersecurity**

MY CAREER VISION

My career goal is to become a Cybersecurity Professional while also exploring the combination of Software and Business. I am interested in using technology not only to solve problems, but also to create useful and innovative solutions.

PERSONAL QUALITY

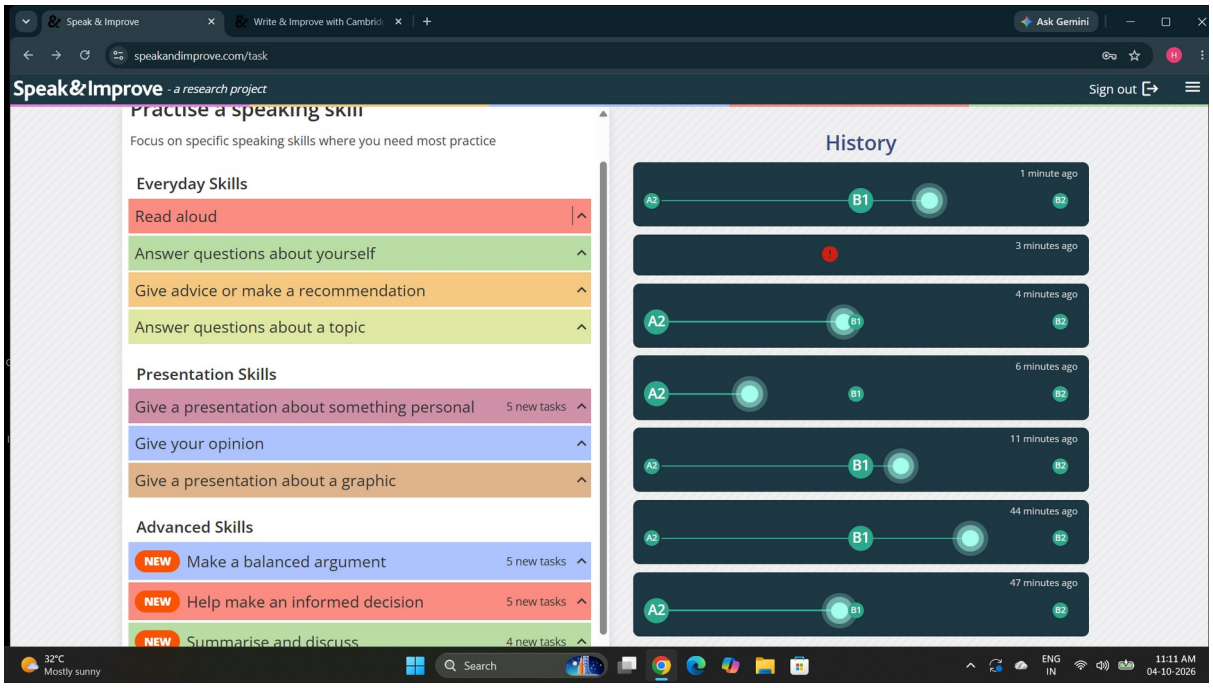
I like combining creativity with technology.

This reflects my interest in bringing together technical knowledge, creative thinking and practical problem-solving.

03 CONCLUSION

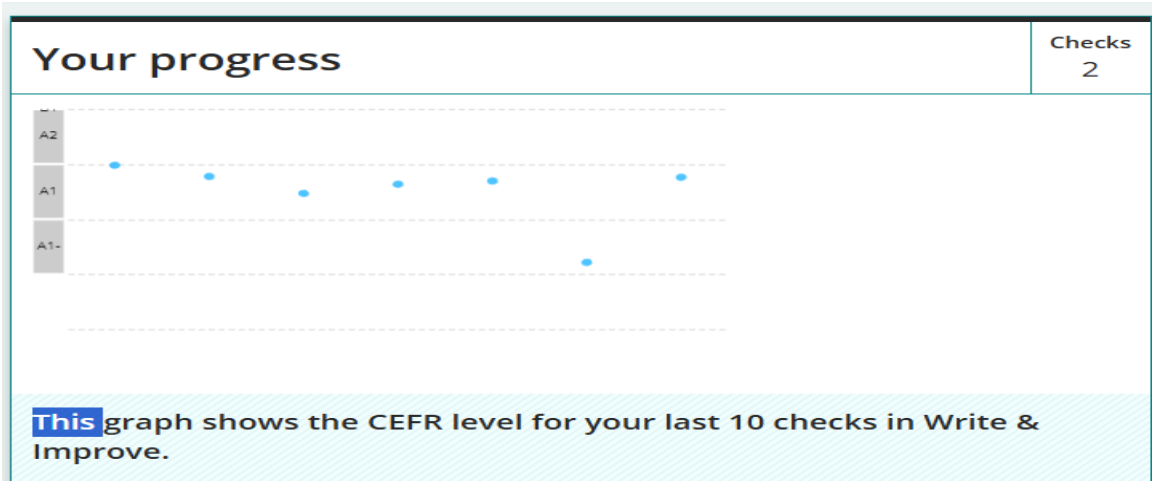
My SWOC analysis helps me understand my strengths, weaknesses, opportunities, and challenges. I am a quick learner who is patient, creative, curious, and comfortable working in teams. However, I need to improve my stage fear, public speaking, overthinking, and ability to stay focused. I have opportunities to develop my technical and communication skills by learning programming languages such as C, using online resources, participating in hackathons and ideathons, and learning from teachers and seniors. My key challenges include balancing academics with skill development, gaining practical experience, completing projects on time, communicating with new people, and choosing the right career path. I am interested in programming, technology, AI, problem-solving, creativity, business, and entrepreneurship. My goal is to become a cybersecurity professional while exploring software and business. I believe my ability to combine creativity with technology will help me grow and achieve my goals.

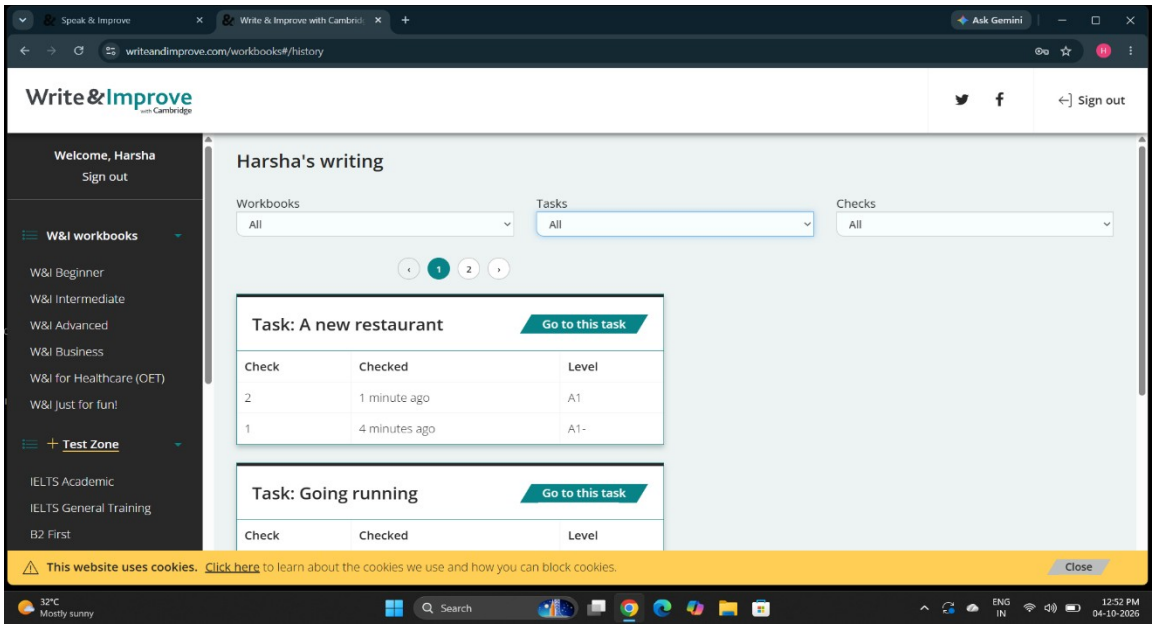
SPEAK AND IMPROVE

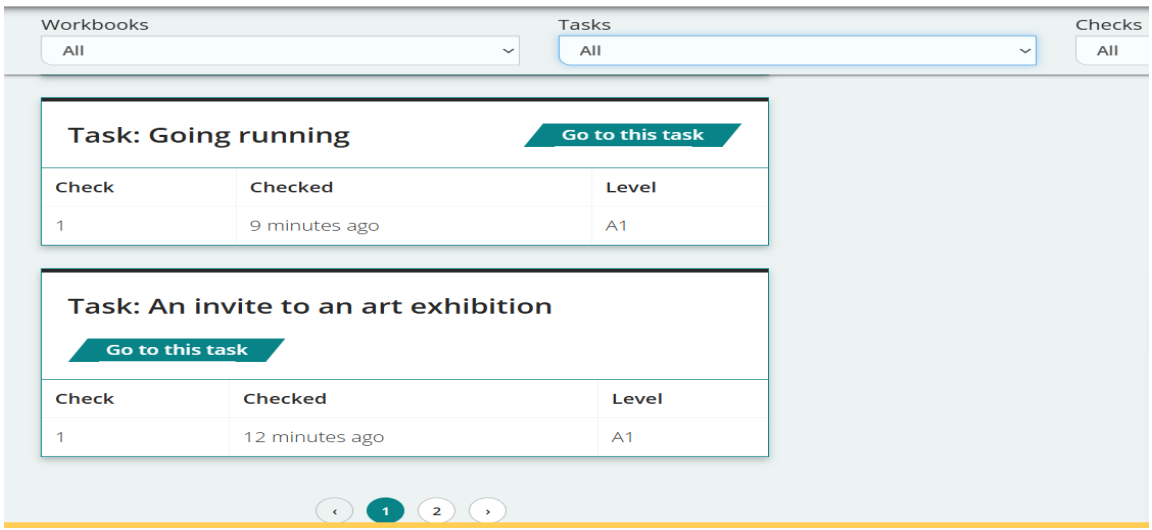


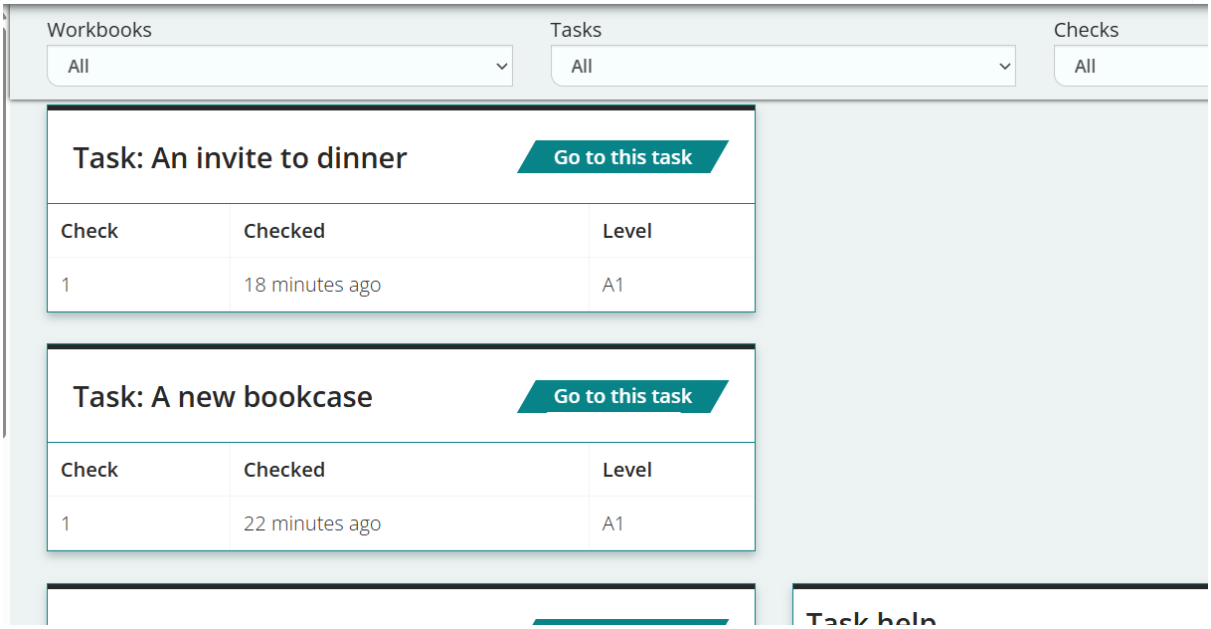


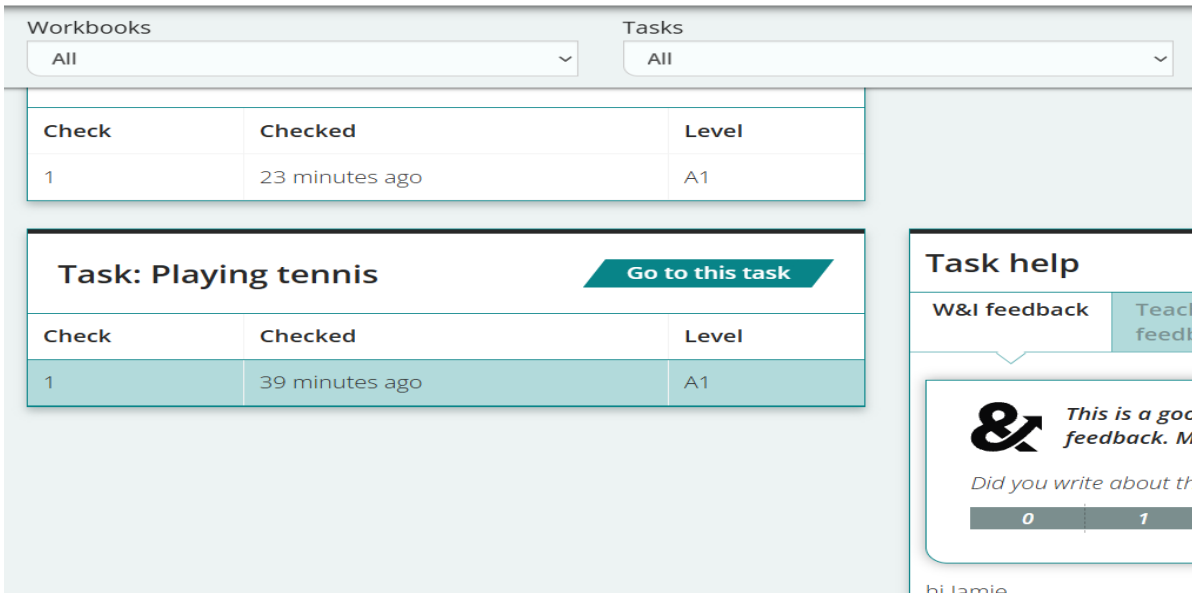
WRITE AND IMPROVE








Workbooks
All
Tasks
All
Checks
All
Task: An invite to dinner
Go to this task
Check
Checked
Level
1
18 minutes ago
A1
Task: A new bookcase
Go to this task
Check
Checked
Level
1
22 minutes ago
A1


Workbooks
All
Tasks
All
Check
Checked
Level
1
23 minutes ago
A1
Task: Playing tennis
Go to this task
Check
Checked
Level
1
39 minutes ago
A1
Task help
W&I feedback
This is a goo
feedback. M
Did you write about th
0
1

ENHANCING SOFT SKILLS AND PERSONALITY

Comprehensive 8-Week Course Synthesis & Executive Summary

INSTRUCTOR
Prof. T. Ravichandran

INSTITUTION
IIT Kanpur (NPTEL MOOC)

STRUCTURE
8 Weeks / 40 Modules

CURRICULUM ARCHITECTURE

Week	Core Theme	Primary Learning Outcomes
Week 1	Mindset Foundations	Growth vs. fixed mindsets, experiential learning, holistic growth.
Week 2	Priority & Time Management	Pareto principle, conquering procrastination, early assertiveness.
Week 3	Emotional Intelligence	Rights of self-assertion, anger regulation, authentic potential.
Week 4	Interpersonal Dynamics	Likeability factors, inner authenticity, self-reflective psychology.
Week 5	Masterful Communication	Eliminating linguistic errors, vocabulary precision, practical fluency.
Week 6	Humor & Financial Wisdom	Constructive workplace levity, non-material value, financial prudence.
Week 7	Physical Well-being & Love	Proactive health maintenance, sleep discipline, balanced relationships.
Week 8	Ethics, Cognition & Planet	Professional ethics, memory enhancement, environmental awareness.

DETAILED WEEKLY MODULES (1 – 3)

Week 1: Mindset Foundations & Behavioral Learning

Units 1–5

- **Action-Oriented Learning:** Education is defined as true personal transformation manifested in observable behavioral changes rather than passive credential gathering.
- **Dweck's Mindset Model:** Contrasts rigid fixed mindsets with expansion-focused growth mindsets, showing that intelligence and capabilities evolve through structured practice.
- **Internal Dialogue:** Recommends eradicating negative self-talk, resisting the urge to inflate daily difficulties, and favoring incremental, continuous self-improvement.
- **Holistic Integration:** Emphasizes balanced progress across physical, psychological, emotional, social, cultural, and spiritual spheres.

Week 2: Priority Allocation & Defeating Delay

Units 6–10

- **Time Allocation:** Frames time as a non-renewable asset; advocates systematically logging activity metrics

DETAILED WEEKLY MODULES (3 – 5)

Week 3: Advanced Assertiveness & Emotional Mastery

Units 11–15

- **Assertive Equilibrium:** Establishes boundary-setting between passive submissiveness and hostile aggression, anchored by a personal Bill of Rights.
- **Hierarchy Drilling:** Advises practicing clear, respectful communication across varied power dynamics (peers, supervisors, and strangers).
- **Emotional Regulation:** Shows how controlling anger yields long-term physical well-being, stable leadership, and higher trust.
- **Reframing Fear:** Synchronizes physical posture, intentional breathing, and constructive thought patterns to turn anxious tension into positive energy.
- **Actualization over Image:** Draws on Bruce Lee's philosophy to urge learners to invest energy into genuine self-building rather than performative outward projection.

Week 4: Interpersonal Appeal & Practical Psychology

Units 16–20

- **Self-Audit of Unpleasant Traits:** Identifies repelling behaviors—gossip, intrusiveness, boundary violations, and grandstanding—for rigorous personal evaluation.
- **Jungian Reflection:** Applies Carl Jung's insight that irritation with external behaviors often illuminates unaddressed aspects of our inner character.
- **Cultivating Appeal:** Highlights core soft skills: active empathy, honoring physical personal space, and leaving every contact on a positive note.
- **Unforced Authenticity:** Frames genuine magnetism as effortless self-trust rather than needy attention-seeking.
- **Law of Attractiveness:** Cites Wayne Dyer to reiterate that individuals draw social responses that reflect what they embody within.

Week 5: Precision in English Expression

Units 21–25

- **Systematic Error Elimination:** Targets frequent errors in vocabulary, grammatical structure, localized idioms, and improper word pairings.
- **Distinction Testing:** Trains precision with easily confused terms (e.g., *personnel* vs. *personal*, *alternate* vs. *alternative*) through active self-testing.
- **Worldview Expansion:** Applies Wittgenstein's idea that broadening language precision expands the boundaries of one's professional and personal opportunities.
- **Practical Applied Mastery:** Promotes direct application and active correction over memorization, leaning on insights from Mark Twain and Richard Branson.
- **Linguistic Immersion:** Encourages listening to high-caliber public speakers and incorporating clear, vivid language into daily conversations.

DETAILED WEEKLY MODULES (6 – 8)

Week 6: Workplace Humor & Financial Balance

Units 26–30

- **Constructive Levity:** Identifies humor as an effective stress-mitigation strategy that builds team cohesion and maintains perspective under strain.
- **Creative Problem-Solving:** Cites evidence showing lighthearted humor relaxes rigid thinking and sparks creative solutions in work environments.
- **Uplifting Communication:** Rejects sarcasm or humor that demeans others, advocating for levity that raises group morale.
- **Non-Material Wealth:** Reinforces that money cannot purchase inner peace, integrity, respect, or authentic personal character.
- **Financial Discipline:** Combines Shakespearean advice against reckless lending/borrowing with practical budget management principles.

Week 7: Vitality, Sleep Discipline & Healthy Dynamics

Units 31–35

- **Holistic Vitality:** Treats physical health as foundational to personality; links self-respect directly to consistent body care and daily habit control.
- **Prevention Ownership:** Highlights personal accountability for health, pointing out that preventive care is far superior to reacting to illness.
- **Dietary & Sleep Rigor:** Links sleep deprivation to cognitive decline and stresses restraint in daily diet.
- **Interdependent Relationships:** Examines personal relationships as growth accelerators, warning against self-neglect or codependency.
- **Paying Kindness Forward:** Encourages balancing self-reliance with supportive community involvement while maintaining personal development goals.

Week 8: Ethics, Memory Expansion & Eco-Stewardship

Units 36–40

- **Ethical Integrity:** Defines genuine refinement as absolute consistency between private values and public conduct.
- **Corporate Conduct:** Outlines professional communication rules, business etiquette, and subtle social protocols.
- **Cognitive Conditioning:** Treats memory and focus as muscle groups requiring active exercise, decluttering, and deliberate training.
- **Retention Techniques:** Applies visualization, structured association, and spaced repetition to retain knowledge over time.
- **Environmental Responsibility:** Concludes by extending soft skills to ecological awareness, identifying care for the planet as the mark of a mature personality.

Ideation Plan

Topic Idea: "Spam Call Detection: Protecting Users from Unwanted and Fraudulent Calls"

Target Angle: Exploring how a first-year student can use programming and basic machine learning to solve an everyday problem that affects millions of phone users in India.

Detailed Outline

1. Introduction (The Hook)

- **Core Concept:** Almost everyone has picked up a call about a fake loan offer, a lottery prize, or a bank KYC update, only to realise it was spam or a scam.
- **Goal:** Capture interest by showing how common these calls are and how much money and time people lose to them, then pose the question: can a simple program tell a spam call from a genuine one before the user answers?

2. Body (The Technical Core)

- **Problem Statement:** Build a system that classifies an incoming call or number as spam or genuine using call patterns and number features.
- **Approach:** Collect or use a sample dataset of numbers with labels (spam/not spam), clean the data, and extract features such as number prefix, call frequency, call duration, time of day, and user spam reports.
- **Detection Logic:** Start with a rule-based method (blacklist, repeated-call checks, suspicious prefixes), then extend to a simple machine learning classifier such as Naive Bayes or Decision Tree in Python.
- **Tools and Technologies:** Python, Pandas, scikit-learn, Git/GitHub for version control, and a simple Tkinter or web interface where a user enters a number and sees the result.
- **Key Features:** Number lookup, spam score with a short reason, user reporting to grow the spam list, and a history log of checked numbers.
- **Challenges:** Limited real data, privacy of phone numbers, false positives that block genuine calls, and scammers frequently changing numbers.

3. Conclusion (The Vision)

- **Expected Outcome:** A working prototype that flags suspicious numbers with a reasonable accuracy and explains why.
- **Future Scope:** Convert the prototype into a mobile app, add real-time call screening, detect scam patterns from voice or call transcripts, and support regional languages such as Tamil.
- **Call to Action:** Build the prototype step by step, test it with real examples, and improve it so that the project can help ordinary people stay safe from phone fraud.

Professional English – I (Lab Integrated)
Model I Practice Worksheet

PART A – GRAMMAR EXERCISES

Question: Ex. 1: Choose the correct subject-verb form: Neither the project supervisor nor the lead programmers ___ the anomaly.

Answer: recognize

Question: Ex. 1: Each of the automated sorting arms ___ recalibration.

Answer: requires

Question: Ex. 1: Either the primary sensor or the auxiliary telemetry units ___ packet data.

Answer: transmit

Question: Ex. 1: The consortium of aerospace contractors ___ to the revised tolerance levels.

Answer: agrees

Question: Ex. 1: Physics and material science ___ the theoretical foundation.

Answer: form

Question: Ex. 2: The metallurgy division ___ fatigue testing last month (conduct).

Answer: conducted.

Question: Ex. 2: The field engineers ___ thirty IoT nodes (deploy).

Answer: deployed.

Question: Ex. 2: The relief valve ___ at 150 psi (open).

Answer: opened.

Question: Ex. 2: The research unit ___ its initial benchmark results (publish).

Answer: published.

Question: Ex. 2: The algorithm ___ all dynamic road obstacles (detect).

Answer: detected.

Question: Ex. 3: GaN semiconductors operate at ___ power efficiency than silicon transistors (high).

Answer: higher.

Question: Ex. 3: Solid-state batteries deliver a ___ operational threshold (safe).

Answer: safer.

Question: Ex. 3: The revised compiler executes ___ (fast).

Answer: faster.

Question: Ex. 3: Brushless DC motors are ___ than brushed equivalents (durable).

Answer: more durable.

Question: Ex. 3: The cooling loop maintained a ___ core temperature (low).

Answer: lower.

Question: Ex. 4: Every operator ___ wear anti-static wrist straps.

Answer: must.

Question: Ex. 4: Hazardous chemical waste ___ be neutralized before disposal.

Answer: must.

Question: Ex. 4: Field technicians ___ log their entry time without fail.

Answer: have to.

Question: Ex. 4: Calibration logs ___ be audited biannually.

Answer: should.

Question: Ex. 4: The test rig ___ undergo an emergency stop inspection.

Answer: must.

Question: Ex. 5: With the upgraded DSP chip, the system ___ process 4K video in real time.

Answer: can.

Question: Ex. 5: If thermal throttling occurs, the clock rate ___ drop by 30%.

Answer: might.

Question: Ex. 5: Laser profilometry ___ measure sub-micron surface roughness.

Answer: can.

Question: Ex. 5: Unshielded signal lines ___ pick up electromagnetic interference.

Answer: may.

Question: Ex. 5: The modular chassis ___ accommodate four expansion cards.

Answer: can.

Question: Ex. 6: Combine: A civil engineer designed the bridge. She received an award.

Answer: The civil engineer who designed the suspension cable bridge received the national infrastructure award.

Question: Ex. 6: Combine: The spectrometer analyzes spectra. It is in the vacuum chamber.

Answer: The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.

Question: Ex. 6: Combine: The specialist was reassigned. His patent was approved.

Answer: The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.

Question: Ex. 6: Combine: The microcontroller has sleep states. It is ideal for remote sensors.

Answer: This microcontroller, which features ultra-low-power sleep states, is ideal for battery-operated remote sensors.

Question: Ex. 6: Combine: Researchers surveyed the farm. They submitted an assessment.

Answer: The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Question: Ex. 7: If the bus voltage ___ 24V, the diode will divert the surge (exceed).

Answer: exceeds.

Question: Ex. 7: The timer will reset the CPU if the thread ___ (fail).

Answer: fails.

Question: Ex. 7: If the operator ___ the safety latch, the press will not actuate (engage).

Answer: engages.

Question: Ex. 7: If temperatures ___ above 90°C, the fans will switch on (rise).

Answer: rise.

Question: Ex. 7: The system will alarm if the detector ___ more than 20 ppm (register).

Answer: registers.

Question: Ex. 8: If the backup generator ___ immediately, the servers would not have shut down (start).

Answer: had started.

Question: Ex. 8: The bearing would not have seized if the crew ___ it (lubricate).

Answer: had lubricated.

Question: Ex. 8: If the team had patched the exploit, the breach ___ (prevent).

Answer: would have been prevented.

Question: Ex. 8: The circuit breaker ___ safely if the sensor had worked (trip).

Answer: would have tripped.

Question: Ex. 8: If the link ___ stable, engineers would have corrected the trajectory (remain).

Answer: had remained.

Question: Ex. 9: Mechanics installed springs. They wanted to isolate resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

Question: Ex. 9: Technicians purge the fuel line. They want to remove vapors.

Answer: Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.

Question: Ex. 9: We introduced an encoder. We intended to determine spindle velocity.

Answer: We introduced an optical encoder to determine spindle velocity accurately.

Question: Ex. 9: Developers wrote a load-testing script to simulate connections.

Answer: The developers wrote an automated load-testing script to simulate ten thousand concurrent connections.

Question: Ex. 9: Gold plating is applied. Its purpose is to prevent corrosion.

Answer: Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Question: Ex. 10: improved fuel economy by ___ the winglet profile (redesign).

Answer: redesigning.

Question: Ex. 10: disconnect before ___ components (replace).

Answer: replacing.

Question: Ex. 10: praised for ___ emissions (reduce).

Answer: reducing.

Question: Ex. 10: Avoid ___ lenses with abrasive solvents (clean).

Answer: cleaning.

Question: Ex. 10: isolates signals by ___ background noise (filter).

Answer: filtering.

Question: Ex. 11: Convert to impersonal passive: The lab assistant poured acid into the flask.

Answer: It is reported that 50 mL of hydrochloric acid was poured into the titration flask.

Question: Ex. 11: The automation unit tested the robotic arm.

Answer: It is reported that the robotic arm was tested under a 50 kg dynamic payload.

Question: Ex. 11: Field engineers calibrated the pyrometer.

Answer: It is reported that the optical pyrometer was calibrated before ignition.

Question: Ex. 11: Technicians observed cavitation damage.

Answer: It was observed that severe cavitation damage was present along the impeller blades.

Question: Ex. 11: We store volatile solvents in cabinets.

Answer: It is a standard practice that volatile solvents are stored in flame-resistant containment cabinets.

Question: Ex. 12: The motor rotates ___ (precise).

Answer: precisely.

Question: Ex. 12: The station ___ polls the sensors (periodic).

Answer: periodically.

Question: Ex. 12: The camera registered the drop ___ (rapid).

Answer: rapidly.

Question: Ex. 12: Joints must be welded ___ (firm).

Answer: firmly.

Question: Ex. 12: The drone traversed the corridor ___ (smooth).

Answer: smoothly.

Question: Ex. 13: Report: “Has the test been documented?”

Answer: The quality manager asked the technician whether the insulation resistance test had been documented.

Question: Ex. 13: Report: “Can the server handle double throughput?”

Answer: The lead architect enquired whether the server cluster could handle double the transaction throughput.

Question: Ex. 13: Report: “Did you inspect the relief lines?”

Answer: The field auditor asked the safety engineer whether he or she had inspected the gas leakage relief lines.

Question: Ex. 13: Report: “Will the update fix the battery drain?”

Answer: The client asked the team whether the firmware update would fix the intermittent battery drain.

Question: Ex. 13: Report: “Why did the multimeter show fluctuating readings?”

Answer: The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Question: Ex. 14: Correct the noun forms: several new machineries and equipments.

Answer: The procurement officer purchased several new machines and pieces of equipment for the workshop.

Question: Ex. 14: Correct: a huge amount of test datas.

Answer: The laboratory generated a huge amount of test data during the wind tunnel trials.

Question: Ex. 14: Correct: three softwares licenses.

Answer: We need three more software licences to complete the finite element analysis on the model.

Question: Ex. 14: Correct: numerous valuable informations.

Answer: The field report provided numerous valuable pieces of information regarding the soil composition.

Question: Ex. 14: Correct: apparatuses and glasswares.

Answer: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Question: Ex. 15: Heat dissipation was ___ than expected (bad).

Answer: worse.

Question: Ex. 15: The antenna achieved a ___ signal-to-noise ratio (good).

Answer: better.

Question: Ex. 15: The vehicle travelled 50 km ___ on one charge (far).

Answer: farther.

Question: Ex. 15: Queues resulted in ___ CPU idle time (little).

Answer: less.

Question: Ex. 15: We require ___ investigation into the cracks.

Answer: further.

PART B – SENTENCE WRITING AND TRANSFORMATION

Question: 1. Rewrite in Simple Past: The technician inspects the solar arrays and records the value.

Answer: The technician inspected the solar arrays and recorded the peak power conversion value.

Question: 2. Rewrite in Simple Past: Engineers deploy water-monitoring nodes.

Answer: The field engineers deployed thirty IoT water-monitoring nodes across the river basin.

Question: 3. Rewrite in Simple Past: The relief valve opens at 150 psi.

Answer: The relief valve opened at 150 psi as designed.

Question: 4. Rewrite in Simple Past: The research unit publishes benchmark results.

Answer: The research unit published its initial benchmark results on low-power RISC-V architectures.

Question: 5. Rewrite in Simple Past: The algorithm detects road obstacles.

Answer: The autonomous navigation algorithm detected all dynamic road obstacles successfully.

Question: 6. Cloud storage vs on-premise servers – cost and scalability.

Answer: Cloud storage usually has a lower initial deployment cost than on-premise servers. It is also more scalable because storage can be increased when required.

Question: 7. Fibre optic vs copper cables – bandwidth and interference.

Answer: Fibre-optic cables provide higher bandwidth than copper cables. They are also less affected by electromagnetic interference.

Question: 8. EV vs ICE vehicles – emissions and price.

Answer: Electric vehicles produce lower tailpipe emissions than ICE vehicles. However, their initial purchase price may be higher than that of conventional vehicles.

Question: 9. Hydraulic vs electromechanical actuators.

Answer: Hydraulic actuators can produce greater force than many electromechanical actuators. Electromechanical actuators have a lower risk of fluid leakage.

Question: 10. SQL vs NoSQL databases.

Answer: Relational databases generally have a more rigid schema than document databases. NoSQL databases can offer more flexible horizontal scaling for suitable workloads.

Question: 11. Combine: The algorithm analyzes cracks. It was patented by the team.

Answer: The algorithm, which was patented by the aeronautics team, analyzes microscopic cracks in turbine blades.

Question: 12. Combine: The engineer authorized shutdown. She noticed a pressure spike.

Answer: The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.

Question: 13. Combine: The laboratory houses a generator. Tests are conducted there.

Answer: The laboratory where high-voltage insulator puncture tests are conducted houses a 400 kV impulse voltage generator.

Question: 14. Combine: We engaged a consultant. His audit exposed ports.

Answer: We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.

Question: 15. Combine: The system uses a neural network. It predicts insulation life.

Answer: The supervisory system utilizes an artificial neural network that predicts transformer insulation life based on dissolved gas concentrations.

Question: 16. Complete: If operating temperature exceeds 85°C, ...

Answer: If the operating temperature exceeds 85°C, the cooling system will start automatically.

Question: 17. Complete: If the operator had pressed the emergency switch, ...

Answer: If the operator had pressed the emergency shutdown switch immediately, the equipment would not have been damaged.

Question: 18. Complete: If supply voltage drops below 200V, ...

Answer: If the supply voltage drops below 200V, the backup power system will activate.

Question: 19. Complete: If the administrator had enabled MFA, ...

Answer: If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.

Question: 20. Complete: If carbon monoxide exceeds 50 ppm, ...

Answer: If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the warning alarm will sound and the ventilation system will start.

Question: 21. Complete: If the crew had checked lubrication, ...

Answer: If the mechanical team had checked the bearing lubrication schedule, the bearing failure would have been avoided.

Question: 22. Complete: If an AGV detects an obstacle within 1 metre, ...

Answer: If the AGV detects an obstacle within 1 metre, it will stop automatically.

Question: 23. Complete: If the thermal sensor had detected the reaction, ...

Answer: If the thermal sensor had registered the runaway reaction in time, the emergency cooling system would have been activated.

Question: 24. Complete: If software receives an unauthenticated API token, ...

Answer: If the software receives an unauthenticated API token, it will reject the request.

Question: 25. Complete: If inspectors had reviewed the core samples, ...

Answer: If the civil inspectors had reviewed the core sample density tests, the structural weakness would have been identified.

Question: 26. Impersonal passive: Chemical engineers distilled the extract at 60°C.

Answer: It was reported that the crude extract was distilled at a constant temperature of 60°C.

Question: 27. Impersonal passive: The metallurgist polished the weld seam.

Answer: It was reported that the cross-section of the weld seam was polished with fine diamond paste.

Question: 28. Impersonal passive: QC rejected fifteen pump casings.

Answer: It was reported that fifteen cast-iron pump casings were rejected due to internal porosities.

Question: 29. Impersonal passive: The technician measured resistance.

Answer: It was reported that the resistance across the bridge circuit was measured using a calibrated digital ohmmeter.

Question: 30. Impersonal passive: The flight crew initiated braking flaps.

Answer: It was reported that the emergency aerodynamic braking flaps were initiated at an altitude of 3,000 metres.

Question: 31. Report: “Did you verify the grounding connections?”

Answer: The safety supervisor asked the intern whether he or she had verified the grounding connections on the generator.

Question: 32. Report: “Has the treatment plant received clearance?”

Answer: The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.

Question: 33. Report: “Why did you disable the firewall rule?”

Answer: The IT security lead enquired of the developer why he or she had disabled the firewall rule on the database server.

Question: 34. Report: “Can the bridge withstand the winds?”

Answer: The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.

Question: 35. Report: “How many trials did you conduct?”

Answer: The research guide asked the postgraduate scholar how many sample trials he or she had conducted under cryogenic conditions.

Question: 36. Correct uncountable nouns: software and machineries.

Answer: We need to install several new software packages and machines in the CAD laboratory.

Question: 37. Correct: useful informations.

Answer: The team collected plenty of useful information from the field test.

Question: 38. Correct: apparatuses and equipments.

Answer: The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.

Question: 39. Correct: infrastructures and electrical powers.

Answer: High-speed train operations require advanced transport infrastructure and a reliable electrical power supply.

Question: 40. Correct: atmospheric datas and pollutions indices.

Answer: The environmental sensor gathers extensive atmospheric data and pollution indices every minute.

Question: 41. Correct: tons of sands, cements.

Answer: The construction crew unloaded tons of sand, cement and steel debris at the worksite.

Question: 42. Correct: hardwares and wirings.

Answer: All electrical hardware and wiring must comply with local safety legislation.

Question: 43. Correct: knowledges.

Answer: The data scientist gained useful knowledge after running the machine-learning clustering.

Question: 44. Correct: laboratory staffs and glasswares.

Answer: Laboratory staff must clear away all glassware and acidic waste before leaving the premises.

Question: 45. Correct: fundings and researches.

Answer: The project failed due to a lack of proper funding and technical research.

Question: 46. Infinitive: The pressure valve vents excess steam.

Answer: A pressure relief valve is installed to vent excess boiler steam.

Question: 47. Gerund: The script archives logs by running at midnight.

Answer: The automated script runs every midnight for archiving legacy server logs.

Question: 48. Infinitive: Engineers installed an intercooler.

Answer: Automotive engineers installed an intercooler to reduce the compressed air intake temperature.

Question: 49. Gerund: Technicians analyzed stress points using gauges.

Answer: The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.

Question: 50. Infinitive: Copper heatpipes conduct heat away.

Answer: High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.

Question: 51. Gerund: The factory reduced water use with recycling.

Answer: The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.

Question: 52. Infinitive: Safety guards prevent contact.

Answer: Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.

Question: 53. Infinitive: The algorithm pre-fetches indices.

Answer: The algorithm optimizes memory bandwidth to pre-fetch sequential array indices into L2 cache.

Question: 54. Infinitive: The UPS protects servers.

Answer: The electronics lab installed a UPS to protect servers against sudden voltage sags.

Question: 55. Gerund: Nitrogen atoms enhanced hardness.

Answer: The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Question: 56. Adverbs: automatic, flawless.

Answer: automatically; flawlessly.

Question: 57. Adverbs: accurate, instant.

Answer: accurately; instantly.

Question: 58. Adverbs: continuous, cautious.

Answer: continuously; cautiously.

Question: 59. Adverbs: consistent, catastrophic.

Answer: consistently; catastrophically.

Question: 60. Adverbs: periodic, smooth.

Answer: periodically; smoothly.

Question: 61. Modals: capability and prohibition – microcontroller drives motors; staff handle reagents.

Answer: can; must not.

Question: 62. Modals: obligation and possibility – personnel wear footwear; condensation form.

Answer: must; might.

Question: 63. Modals: prohibition and past capability.

Answer: must not; could.

Question: 64. Modals: advice and logical deduction.

Answer: should; must.

Question: 65. Modals: capability and prohibition.

Answer: can; must not.

PART C – LONG ANSWERS (MODEL RESPONSES)

Question: Topic 1, Q1: Eight safety instructions for a Chemistry and Materials Testing Laboratory.

Answer: 1. Wear a lab coat, gloves and safety goggles at all times. 2. Read the safety instructions before handling chemicals. 3. Never touch or smell chemicals directly. 4. Use a fume hood when working with harmful vapours. 5. Label every chemical container clearly. 6. Keep food and drinks outside the laboratory. 7. Clean spills immediately by following the approved procedure. 8. Switch off equipment and wash your hands before leaving.

Question: Topic 1, Q2: Eight safety instructions for a High-Voltage Electrical Laboratory.

Answer: 1. Enter the laboratory only with permission. 2. Wear approved insulated protective equipment. 3. Keep hands and tools away from live electrical parts. 4. Switch off and isolate the supply before making connections. 5. Check cables and insulation before every experiment. 6. Keep the work area dry and clean. 7. Do not operate equipment without supervision. 8. Know the location of the emergency switch and follow the instructor’s directions.

Question: Topic 1, Q3: Eight safety instructions for a CNC workshop.

Answer: 1. Wear safety goggles and suitable footwear. 2. Tie back long hair and secure loose clothing. 3. Check the machine guard before operation. 4. Use the machine only after receiving training. 5. Keep hands away from moving cutters and spindles. 6. Remove chips only with an approved brush after the machine stops. 7. Clean coolant spills immediately. 8. Switch off the machine before maintenance or tool changes.

Question: Topic 2, Q1: Harvest festival and eco-friendly practices (about 250 words).

Answer: Pongal is one of the most important harvest festivals celebrated in Tamil Nadu. It is a festival of gratitude to the Sun, nature, farmers and cattle for supporting agriculture. Families clean their homes, draw colourful kolams and prepare the traditional Pongal dish in new pots. The celebration brings relatives and neighbours together and creates a feeling of unity. On the first day, people celebrate Bhogi; on the second day, they offer Pongal to the Sun; Mattu Pongal honours cattle; and Kaanum Pongal is a day for family visits. Traditional meals include sweet Pongal, ven Pongal, sugarcane, vegetables and other homemade dishes. These foods represent the importance of farming and local produce. Modern eco-friendly practices can preserve the festival while protecting nature. People can avoid plastic decorations and use banana leaves, clay pots and reusable plates. Natural materials such as flowers, leaves and rice flour can be used for decoration. Communities should separate food waste and send it for composting. Farmers' markets can promote locally grown products and reduce transport-related pollution. Public celebrations can use energy-efficient lights and solar power. People should also avoid excessive noise and wasteful use of water. Schools and colleges can organize awareness programmes about sustainable farming and traditional culture. Pongal is more than a celebration; it reminds us to respect nature and appreciate farmers. By combining traditional customs with responsible practices, we can pass its spirit to future generations.

Question: Topic 2, Q2: Temple car festival and sustainable practices (about 250 words).

Answer: A traditional temple car festival is a colourful cultural event in many towns of Tamil Nadu. The decorated temple car is taken through the streets with the participation of devotees, musicians and local communities. People gather to offer prayers, pull the car and welcome relatives who visit during the festival. Traditional music, kolams, flower decorations and devotional songs create a special atmosphere. Food stalls serve local dishes, snacks, buttermilk and prasadam. These customs strengthen unity and preserve local heritage. However, large crowds can create waste and noise. Sustainable methods can make the festival cleaner and safer. Organizers should avoid single-use plastic cups, plates and carry bags. Reusable steel tumblers, banana leaves and paper-based materials can be encouraged. Separate bins should be placed at important points for wet and dry waste. Food waste can be collected for composting, while reusable materials can be sent for recycling. Volunteers can guide visitors through marked routes and keep emergency paths clear. Drinking-water stations can reduce the use of plastic bottles. Loudspeaker volume should be controlled, and traditional music can be given importance over unnecessary amplified sound. Solar-powered lights can be used where suitable. Public announcements should encourage visitors to keep the streets clean. Local authorities, temple committees and residents must work together. The temple car festival represents faith, culture and community spirit. Protecting the environment during the celebration ensures that future generations can enjoy this valuable tradition.

Question: Topic 2, Q3: Folk art and handicraft fairs using solar power and composting.

Answer: Regional folk art and handicraft fairs celebrate the creativity and traditions of local communities. Artisans display handmade pottery, woven products, woodwork, paintings, toys and traditional ornaments. Folk music and dance performances make the fair lively. Visitors can also enjoy regional delicacies prepared using local ingredients. Such fairs provide income to artisans and help younger generations understand their cultural heritage. Modern facilities can make these events more sustainable. Organizers can install solar micro-grids to provide electricity for lighting, small stalls and information counters. Energy-efficient LED lamps can reduce power consumption. Reusable decorations made from cloth, bamboo and natural materials can replace plastic banners. Separate waste bins should be placed throughout the venue. Food waste and biodegradable materials can be collected for zero-waste composting. Vendors can serve food in reusable or compostable containers. Visitors should be encouraged to carry cloth bags and refillable water bottles. Digital tickets and online information boards can reduce paper waste. Workshops conducted by experienced artisans can teach traditional skills to students. Local schools and colleges can invite students to participate in demonstrations and cultural programmes. These fairs connect the past with the present. By using renewable energy and responsible waste management, communities can protect the environment while supporting traditional artists and handicrafts.

Question: Topic 3, Q1: Technical study tour to a hydroelectric dam (about 250 words).

Answer: Our college can organize a two-day technical study tour to a major hydroelectric dam. The group will travel by college bus after attendance and safety instructions. The route and travel time will be confirmed by the faculty coordinator. Students will carry identity cards, notebooks, water bottles and basic medicines. On arrival, the group will attend an introductory briefing about the dam and its purpose. The technical visit will include the reservoir, intake structure, penstock, turbine hall, generator units and control room, subject to official permission. Students can learn how stored water is directed towards turbines and how mechanical energy is converted into electrical energy. They can observe the role of transformers, protection systems and monitoring instruments. Faculty members will explain the importance of water management and grid connection. During the journey, students can taste local food such as idli, dosa, rice meals and regional snacks. Buying food from local vendors can also support the nearby community. Students must follow all site rules, stay with the group and avoid restricted areas. They should wear comfortable shoes and use helmets or other protective equipment if provided. Photography must be done only with permission. After the visit, students will prepare a report with diagrams and observations. The tour will connect classroom lessons with real industrial applications and improve our understanding of renewable energy.

Question: Topic 3, Q2: Technical tour to a solar and wind hybrid farm.

Answer: Our department plans a technical study tour to an integrated solar and wind hybrid energy farm. The students and faculty will travel by college bus, following a planned route and timetable. Before departure, the coordinator will confirm entry permission, transport, food and accommodation. Students will carry identity cards, notebooks, water and sun protection. The visit will begin with an introduction to renewable energy and the purpose of combining solar and wind power. At the solar section, students will observe photovoltaic panels, mounting structures, inverters and monitoring systems. The guide will explain how sunlight is converted into electricity and how panel output is monitored. At the wind farm, students will learn about turbine blades, towers, generators and grid connections. They will also understand how wind speed affects power generation. During the journey, the group can enjoy local dishes such as idli, dosa, lemon rice and regional snacks. The trip will include scheduled breaks and a head count at every stop. Students must remain in permitted visitor areas, follow instructions and avoid touching electrical equipment. They should wear suitable footwear and stay away from restricted turbine zones. Photography should be done only with permission. At the end, each student will prepare a short report comparing solar and wind energy. The tour will provide practical knowledge about clean energy technologies and their role in sustainable development.

Question: Topic 3, Q3: Technical visit to a seawater RO desalination plant.

Answer: Our college can arrange a technical visit to an automated seawater reverse osmosis desalination plant. The students will travel by college bus along the coastal route under faculty supervision. The coordinator will arrange permission, transport and refreshments in advance. Students will carry identity cards, notebooks and drinking water. The visit will begin with a safety briefing and an explanation of the need for desalination in water-scarce regions. At the plant, students will observe the seawater intake pumping station, pre-treatment units, high-pressure pumps and membrane filtration racks. Engineers will explain how reverse osmosis membranes separate dissolved salts from water. Students will also learn about post-treatment, water-quality monitoring, storage and distribution. The guide will explain how the plant manages the concentrated brine produced during the process. The group can sample regional coastal food such as idli, dosa, rice meals and locally prepared snacks at an approved stop. Students must stay with the group, wear the safety equipment provided and avoid touching pumps, pipes or control panels. They must follow all restricted-area instructions and take photographs only with permission. After the visit, students will prepare a report on the process, energy use and environmental considerations. This trip will help us connect water-treatment theory with real industrial practice.

Question: Topic 4, Q1: Formal request for after-hours lab access for a National AI Hackathon.

Answer:
T. Harshavardhan
First-Year B.E. CSE
St. Joseph’s College of Engineering
OMR, Chennai - 119

Date: 06 November 2026

Dr. R. Meenakshi
Head, Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai - 119

Subject: Request for after-hours laboratory access for National AI Hackathon preparation

Respected Sir/Madam,

We are a student team preparing to participate in a National AI Hackathon. We request permission to use the computer laboratory after regular college hours for project development and testing. Our objective is to develop an AI-based solution that addresses a practical problem and present a working prototype at the event.

We require computers with internet access and approved software such as Python, code editors, database tools and relevant AI development libraries. We assure you that we will use the facilities responsibly, follow laboratory rules and leave the room clean. We will work under the guidance of a faculty mentor and will not disturb other academic activities.

Kindly grant us permission for the required period. Thank you.

Yours faithfully,
T. Harshavardhan
On behalf of the project team

Question: Topic 4, Q2: Request for an industrial visit for 50 first-year students.

Answer:
Arjun Kumar
Class Representative, First-Year B.E.
St. Joseph's College of Engineering
OMR, Chennai - 119

Date: 06 November 2026

Mr. S. Venkatesh
Plant Manager
Industrial Robotics and Automation Factory
SIPCOT Industrial Park, Sriperumbudur, Kancheepuram - 602105

Subject: Request for permission to conduct an industrial visit for 50 students

Respected Sir/Madam,

We request permission to visit your factory with 50 first-year engineering students and faculty members. The purpose is to understand industrial robotics, automated production lines, sensors, actuators and quality-control systems. The visit will help students connect classroom concepts with practical applications.

We request a suitable date, a technical briefing and a guided tour of the permitted production areas. We will follow your safety rules, wear the required protective equipment and remain with the assigned guides. Kindly inform us about entry procedures and any documents required.

We would be grateful for your approval. Thank you.

Yours faithfully,
Arjun Kumar
Class Representative

Question: Topic 4, Q3: Request for machine shop access for an electric go-kart.

Answer:
Karthik Raman
Project Team Representative, Department of Mechanical Engineering
St. Joseph’s College of Engineering
OMR, Chennai - 119

Date: 06 November 2026

Dr. P. Sundaram
Head, Department of Mechanical Engineering
St. Joseph’s College of Engineering
OMR, Chennai - 119

Subject: Request for after-hours machine shop access

Respected Sir/Madam,

We request permission to use the department machine shop after college hours to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge. The project will help us learn design, fabrication, motor selection, battery integration and testing.

We require supervised access to approved workshop machines, measuring tools and basic fabrication facilities. We assure you that all work will be carried out under faculty supervision and according to workshop safety rules. We will maintain the equipment and clean the workspace after use.

Kindly grant us permission for the required period. Thank you.

Yours faithfully,
Karthik Raman
Project Team Representative

Question: Topic 4, Q4: Request for sponsorship and duty leave for an expo.

Answer:
Divya Lakshmi
Team Representative, Five-Member Student Team
St. Joseph’s College of Engineering
OMR, Chennai - 119

Date: 06 November 2026

Dr. S. Ramanathan
Principal
St. Joseph’s College of Engineering
OMR, Chennai - 119

Subject: Request for financial sponsorship and duty leave

Respected Sir/Madam,

Our five-member multidisciplinary team has been selected to present a prototype at the National Defense Innovation Expo. We request financial support for registration, travel, accommodation and prototype transportation. We also request duty leave for the dates of the event and travel.

Participation will improve our technical knowledge, teamwork, presentation skills and understanding of innovation. We will submit participation proof and a report after returning. We assure you that we will represent the college responsibly.

Kindly consider our request. Thank you.

Yours faithfully,
Divya Lakshmi
Team Representative

Question: Topic 5, Q1: EVs versus conventional ICE vehicles (250–300 words).

Answer: Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two important forms of road transport. Both provide mobility, but they differ in energy source, cost and environmental impact. EVs use electric motors powered by rechargeable batteries, whereas ICE vehicles use petrol or diesel engines. One major advantage of EVs is that they produce no tailpipe emissions during driving. This can improve air quality in cities. In contrast, ICE vehicles release exhaust gases while operating. EVs also have fewer moving engine parts and may require less routine maintenance. Their running cost can be lower when electricity is affordable. However, battery packs are expensive and gradually lose capacity over time. Battery production also requires minerals and energy, creating an environmental footprint. On the other hand, ICE vehicles are widely available and can be refuelled quickly at established fuel stations. Their initial purchase price may be lower in many market segments. However, fuel and maintenance costs can be higher over the long term. Charging infrastructure is another important difference. EV charging networks are expanding, but charging points may be limited in some rural areas and on long routes. Home charging is convenient for many owners, while public fast charging can take longer than refuelling a conventional vehicle. In conclusion, EVs offer cleaner local operation and potentially lower running costs, whereas ICE vehicles provide established refuelling convenience. The better choice depends on travel needs, charging access, purchase budget and electricity sources. Improvements in battery technology and renewable electricity can make EVs more sustainable in the future.

Question: Topic 5, Q2: Remote versus on-site internships (250–300 words).

Answer: Remote and on-site internships both help engineering students gain professional experience. However, they offer different learning environments. Remote internships allow students to work from home or another location using digital tools. On-site internships require students to attend a workplace physically. One major advantage of remote internships is flexibility. Students can save travel time and may manage their academic schedule more easily. They can also work with organizations located in other cities. In contrast, on-site internships provide direct exposure to laboratories, machines, production areas and workplace practices. This is especially useful for engineering students who need hands-on experience. Mentorship is possible in both formats. Remote interns can communicate through video meetings and online collaboration tools, whereas on-site interns may receive immediate guidance from supervisors and colleagues. Face-to-face interaction can make it easier to ask questions and understand practical demonstrations. Networking is another important factor. On-site internships allow students to build personal relationships with professionals and other interns. Remote internships can still provide networking opportunities through online meetings, but informal interaction may be limited. Remote work also requires strong time management and self-discipline. On-site work may involve travel expenses and fixed working hours. Both formats have advantages and limitations. A remote internship is suitable for software development, online research and data-related tasks. An on-site internship is valuable for manufacturing, electronics testing and laboratory work. In conclusion, students should choose an internship based on their learning goals. A combination of remote and on-site activities can provide flexibility as well as practical experience.

Question: Topic 5, Q3: Cloud computing versus on-premise data centres (250–300 words).

Answer: Cloud computing and on-premise data centres are two common ways for enterprises to manage digital infrastructure. Cloud services provide computing resources through the internet, whereas on-premise systems are installed and maintained within an organization. Cloud computing usually requires lower initial capital expenditure because companies do not need to purchase all the physical servers and networking equipment. They can pay according to usage. It is also easier to increase or decrease resources when workload demand changes. In contrast, on-premise data centres require a large initial investment in hardware, buildings, cooling and power systems. However, they provide direct control over equipment and data storage. Security and compliance are important for both models. Cloud providers offer security tools and professional maintenance, but organizations must configure access controls properly and understand where data is stored. On-premise systems allow an organization to manage its own security policies, but the responsibility for updates, backups and protection remains with its IT team. Recurring overheads also differ. Cloud services have ongoing subscription or usage charges. On-premise centres involve electricity, cooling, repairs, staff and hardware replacement. Cloud computing is often more suitable for changing workloads and rapid expansion. On-premise systems may be preferred when special hardware, strict internal control or predictable workloads are required. In conclusion, neither model is perfect for every enterprise. Organizations should compare cost, security requirements, technical skills and scalability before choosing. A hybrid model can combine the flexibility of cloud services with the control of local infrastructure.

Question: Topic 5, Q4: 3D printing versus CNC milling (250–300 words).

Answer: 3D printing and CNC milling are widely used manufacturing methods for producing prototypes. 3D printing is an additive process that builds an object layer by layer. CNC milling is a subtractive process that removes material from a solid workpiece. One major advantage of 3D printing is its ability to create complex shapes, internal channels and customized designs. It generally produces less material waste because material is added only where required. In contrast, CNC milling removes unwanted material, which can create more waste depending on the design. However, CNC milling can produce strong and accurate components from engineering materials such as metals. It often provides a better surface finish and dimensional accuracy for suitable parts. For large batch production, CNC machining can be faster and more consistent once the process is set up. 3D printing may take longer for each part, especially when objects are large or require fine layer resolution. On the other hand, 3D printing needs less special tooling and is useful for quick design changes. CNC milling has geometric limits because the cutting tool must reach the required surfaces. Additive manufacturing can produce more complicated geometries, although support structures and post-processing may be needed. Both methods have limitations in material choice, cost and production speed. In conclusion, 3D printing is suitable for complex, customized and low-volume prototypes. CNC milling is often better for accurate, strong parts and repeat production. Engineers should select the method based on design, material, quantity, finish and budget.

Question: Topic 6, Q1: Personal email to a high-school mentor about first semester.

Answer: To: sathyamoorthy.sir@gmail.com | Subject: My First Semester at Engineering College
Dear Sathyamoorthy Sir,
I hope you are doing well. I am writing to share my experiences during my first semester at engineering college. College life is different from school because I have more independence and responsibility. I have made new friends and learned to manage my daily schedule.
I have attended programming classes and completed basic laboratory activities. Working with classmates has helped me improve my teamwork and communication skills. I am also learning how to prepare lab records and complete assignments on time.
My end-semester examinations are approaching, so I have prepared a study timetable. I revise important topics every day, practise programming questions and clarify doubts with teachers. I am trying to balance theory subjects with practical preparation.
Thank you for your guidance and encouragement. I will always remember the lessons I learned from you.
Yours sincerely,
Harshavardhan

Question: Topic 6, Q2: Personal email to a science teacher about engineering term.

Answer: To: lakshmi.teacher@gmail.com | Subject: Updates from My First Engineering Term
Dear Lakshmi Ma’am,
I hope you are fine. I wanted to share how my first term in engineering college has been. The course is interesting because it includes practical laboratory work along with theory subjects. I have started learning programming and basic engineering concepts through lab activities.
I also got an opportunity to take part in a technical club activity and learn about teamwork and new technologies. These experiences have helped me become more confident and curious. I am slowly adapting to the college timetable and the responsibility of completing assignments and lab records.
For the upcoming university examinations, I am revising class notes, practising important questions and preparing for practical demonstrations. I have made a timetable so that I can give enough time to every subject.
Thank you for inspiring me to learn science. Your teaching has helped me build a strong foundation.
With regards,
Harshavardhan

Question: Topic 6, Q3: Email to elder sibling/alumnus asking for advice.

Answer: To: vignesh.kumar@gmail.com | Subject: Need Your Advice on Managing College Work
Hi Vignesh,
I hope you are doing well. College life has been interesting, but I am finding it challenging to balance academics and technical activities. My day includes classes, laboratory sessions, assignments and time for learning programming. I also want to participate in technical clubs and project activities.
Sometimes, lab record submissions and regular assignments take more time than expected. Because of this, I find it difficult to plan revision and skill development together. I want to improve my technical knowledge without affecting my academic performance.
Could you please suggest a practical way to manage my daily schedule? I would also like to know how you prepared for semester examinations, especially how you revised theory subjects and practised practical questions. Any tips for avoiding last-minute work would be helpful.
Thank you for your guidance. I look forward to hearing from you.
With love,
Harshavardhan

Question: Topic 7, Q1: Four-day technical tour to an aerospace and satellite launch station.

Answer: Day 1: Travel by train or college bus to the selected city. After check-in, attend a briefing about the visit schedule and safety rules. Day 2: Visit the approved visitor centre and attend presentations about satellite design, launch vehicles, communication and mission control. Day 3: Observe permitted exhibition areas and demonstrations about propulsion concepts, payloads, tracking and ground stations. Day 4: Attend a review session and return to college. Estimated per-student budget: travel ■2,000; accommodation ■2,400; food ■1,200; local transport ■500; entry and miscellaneous expenses ■400. Total: ■6,500. The college should confirm actual prices before booking. Students must carry identity cards, follow security instructions, stay with the group and avoid restricted areas. Photography is allowed only where permitted. The tour will provide useful exposure to aerospace engineering and satellite technology.

Question: Topic 7, Q2: Four-day tour to a commercial port and dry-dock facility.

Answer: Day 1: Travel to the port city by train or college bus and check into pre-booked accommodation. Attend a briefing on port operations and safety. Day 2: Visit approved container-terminal areas and learn about cargo handling, cranes, logistics and port management. Day 3: Visit the permitted dry-dock viewing area and attend a session about ship maintenance, hull inspection and maritime engineering. Day 4: Attend a discussion with technical staff and return to college. Estimated per-student budget: travel ■1,800; accommodation ■2,400; food ■1,200; local transport ■500; entry and miscellaneous expenses ■400. Total: ■6,300. Actual costs must be confirmed before the trip. Students must wear helmets, safety shoes and reflective jackets where required. They must remain with the guide, obey restricted-area signs and never approach moving cargo equipment. The tour will help students understand maritime logistics and industrial engineering.

Question: Topic 7, Q3: Four-day tour to locomotive manufacturing and railway signalling centre.

Answer: Day 1: Travel by train to the selected industrial city and check into accommodation. Attend an orientation about the visit plan. Day 2: Visit the approved locomotive manufacturing facility and learn about electric motors, bogies, braking systems and quality testing. Day 3: Visit the railway signalling centre to understand track circuits, signals, control systems and railway safety. Day 4: Conduct a review session, prepare notes and return to college. Estimated per-student budget: rail travel ■1,500; accommodation ■2,400; food ■1,200; local transport ■500; entry and other expenses ■400. Total: ■6,000. The college should confirm the actual costs and permissions. Students must carry identity cards, follow staff instructions, wear required protective equipment and remain within visitor zones. They must not touch machinery or cross railway tracks. The tour will provide practical knowledge about railway manufacturing and signalling technology.

EN25C09: Professional English - I (Lab Integrated)

PART A (15 x 1 = 15 Marks)

1. **Fill in the blank with the correct form of the verb given in bracket:**

   The automated system _____ (monitor) the temperature continuously to ensure safe operation.

   Answer: monitors

2. **Fill in the blank with the correct past form of the verb given in bracket:**

   The engineers _____ (record) the vibration readings during yesterday's trial.

   Answer: recorded

3. **Complete the sentence with the appropriate comparative form of the adjective in bracket:**

   The newly developed composite material is significantly _____ (strong) than the conventional material while maintaining a lower weight.

   Answer: stronger

4. **Fill in the blank with an appropriate modal auxiliary verb: (must/can/might)**

   Engineers _____ follow safety regulations while operating heavy machinery.

   Answer: must

5. **Fill in the blank with a suitable modal auxiliary verb expressing capability: (can, may, might)**

   Modern sensors _____ detect even minor changes in temperature.

   Answer: can

6. **Combine the sentences into a single coherent sentence using a suitable relative pronoun:**

   (a) The laboratory developed a new testing device. (b) The device measures material strength with high precision.

   Answer: The laboratory developed a new testing device **that** measures material strength with high precision.

7. **Complete the sentence using the correct form of the verb in bracket (First Conditional - If Clause):**

   If the temperature _____ (rise) above the permissible limit, the cooling system will activate automatically.

   Answer: rises

8. **Complete the sentence using the correct form of the verb in bracket (Third Conditional - Past Unreal Conditional):**

   If the technician had inspected the circuit earlier, the equipment _____ (not / fail) during operation.

   Answer: would not have failed

9. **Combine the following sentence using an infinitive of purpose (to + verb):**

   The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

   Answer: The mechanics installed dual damper springs **to isolate chassis resonance**.

10. **Fill in the blank with the gerund form of the verb in bracket after the preposition:**

    The research team reduced energy consumption by _____ (optimize) the motor control algorithm.

    Answer: optimizing

11. **Convert the following active voice into the Impersonal Passive Voice:**

    The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

    Answer: 50 mL of hydrochloric acid was poured into the titration flask.

12. **Fill in the blank with the appropriate adverbial form derived from the adjective in bracket:**

    The robotic arm moves _____ (smooth) to ensure accurate positioning of the component.

    Answer: smoothly

13. **Rewrite the direct into indirect speech:**

    The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

    Answer: The project supervisor asked the engineer **whether he had calibrated the pressure sensor before testing.**

14. **Identify the incorrect noun form and rewrite the sentence correctly:**

    The laboratory purchased several new equipments for conducting material testing.

    Answer: Incorrect word: *equipments* (uncountable noun).
    Correct sentence: The laboratory purchased several **pieces of** new **equipment** for conducting material testing.

15. **Fill in the blank using the correct irregular comparative form (better, worse, farther/further, less, more):**

    Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

    Answer: worse

PART B (10 x 2 = 20 Marks)

1. **Rewrite the following active operational and laboratory description in the Simple Past Tense:**

   The engineer calibrates the pressure sensor and verifies its accuracy.

   Answer: The engineer **calibrated** the pressure sensor and **verified** its accuracy.

2. **Draft two comparative sentences comparing a laptop and a desktop computer in terms of portability and screen size.**

   Answer: Portability: A laptop is more portable than a desktop computer.
   Screen size: A desktop computer usually has a larger screen than a laptop.

3. **Combine the technical statements into a single cohesive sentence using a defining or non-defining relative clause:**

   The engineer designed a new cooling system. It reduces the temperature of industrial machines.

   Answer: The engineer designed a new cooling system, **which reduces the temperature of industrial machines**.

4. **Complete the following conditional sentences logically:**

   (a) If the battery voltage drops below the required level, _____

   (b) If the technician had checked the connections earlier, _____

   Answer: (a) If the battery voltage drops below the required level, **the system will shut down automatically / an alarm will sound.**
   (b) If the technician had checked the connections earlier, **the fault would have been detected and the breakdown avoided.**

5. **Convert the following active sentence into the Impersonal Passive Voice suitable for formal engineering reports:**

   The laboratory assistant heated the solution to 80°C before testing.

   Answer: The solution **was heated** to 80°C before testing.

6. **Rewrite the following direct engineering query into Reported (Indirect) Speech:**

   The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

   Answer: The project manager asked the technician **whether he had inspected the pressure valves before starting the compressor.**

7. **Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:**

   (a) The engineer received many feedbacks on the new design.

   (b) The students received several useful advices from their teacher.

   Answer: (a) The engineer received **a lot of feedback** on the new design. ("Feedback" is uncountable, so no -s.)
   (b) The students received several **pieces of useful advice** from their teacher.

8. **Rewrite the sentences using an Infinitive of Purpose or a Gerund construction:**

   (a) The technician wears safety gloves. He wants to protect his hands.

   (b) The cooling fan operates continuously. It prevents the motor from overheating.

   Answer: (a) The technician wears safety gloves **to protect his hands.**
   (b) The cooling fan operates continuously **to prevent the motor from overheating.**

9. **Fill in the blanks with the correct adverbial forms of the bracketed words:**

   (a) The machine operates _____ (efficient) to reduce energy consumption.

   (b) The technician checked the electrical connections _____ (careful) before starting the machine.

   Answer: (a) **efficiently**
   (b) **carefully**

10. Insert suitable modal auxiliary verbs to match the specified functional meaning:

(a) The robotic arm _____ (capability) lift heavy components with precision.

(b) Visitors _____ (strict prohibition) enter the testing area without authorization.

Answer: (a) **can**
(b) **must not**

PART C (5 x 13 = 65 Marks)

Question 26

26. a) Write a set of eight clear, imperative safety instructions to promote safe and responsible use of the internet and protect users from cyber threats. (13 Marks)

Answer:

1. Use strong, unique passwords and change them regularly.
2. Enable two-factor authentication on all important accounts.
3. Do not click on suspicious links or open unknown email attachments.
4. Never share personal details, OTPs or banking information online.
5. Keep your operating system, browser and antivirus software updated.
6. Use only secure websites that begin with "https" for transactions.
7. Avoid public Wi-Fi for banking, or use a VPN.
8. Report cyberbullying, phishing and fraud to the cybercrime portal immediately.

Question 27

27. a) Write a descriptive essay of 250 words about "A Traditional art form in your region and how modern initiatives can preserve its cultural significance." Discuss its historical origins, traditional practices, and ways to promote and sustain it among younger generations. (13 Marks)

Answer (27.a)

In Tamil Nadu, the day begins with kolam, the intricate rice-flour designs drawn at the threshold of homes. More than decoration, this art form carries centuries of cultural meaning.

Kolam has roots in ancient Tamil tradition and is mentioned in early Tamil literature. Women rise before dawn, sprinkle water on the ground and draw geometric patterns, either dots (*pulli kolam*) or free-flowing curves. Coarse rice flour was traditionally used so that ants and birds could feed on it, reflecting the belief in living in harmony with nature. During Margazhi and Pongal, the designs grow larger and are filled with colour. The patterns also encode mathematical ideas such as symmetry, loops and fractals.

Urban living, apartment floors and busy schedules have pushed kolam toward stickers and plastic stencils. Modern initiatives can reverse this. Schools and colleges can hold kolam competitions and workshops. Digital platforms can host tutorials and pattern libraries. Researchers can document designs, and artists can adapt them for fabrics, murals and product design. Cultural festivals and social media challenges can give young people recognition and a sense of pride. Using eco-friendly materials keeps the practice sustainable. Families, too, can help by drawing kolam together every morning, so that children learn the patterns and their meaning naturally.

Kolam is a living heritage that blends art, science and spirituality. If young people are encouraged to see it as creative expression rather than a chore, this ephemeral art form will continue to brighten Tamil homes for generations.

27. b) Write a structured travel plan and descriptive feature (250 words) on "An Educational Trip to a Place of Historical and Cultural Importance." Include travel arrangements, important landmarks, local food experiences, and observations of the region's cultural heritage and traditions. (13 Marks)

Answer (27.b)

Introduction: Our department organised a one-day educational trip to Mahabalipuram, a UNESCO World Heritage Site about 60 km from Chennai, to see the architectural brilliance of the Pallava dynasty at first hand.

Travel arrangements: Sixty students and four faculty members travelled in two air-conditioned buses, leaving the college at 6:30 a.m. along the scenic East Coast Road and returning by 7:00 p.m. Tickets, permissions and refreshments were arranged in advance by the student committee, and a first-aid kit accompanied the group. Students were asked to carry notebooks, water bottles and their college ID cards, and to follow the instructions of the faculty coordinators throughout the visit.

Important landmarks: We first visited the Shore Temple, a granite structure standing beside the Bay of Bengal. We then explored the Five Rathas, monolithic temples carved from single rocks, followed by Arjuna's Penance, a huge relief sculpture telling stories from the epics. A local guide explained the history and engineering skill behind each monument, pointing out how the Pallava artists planned their carvings without modern tools.

Local food experiences: At lunch we enjoyed a traditional Tamil meal served on banana leaves, including sambar, rasam, kootu and payasam, and tasted fresh seafood specialities from the coastal stalls, which gave us a feel for the region's everyday culture.

Observations and conclusion: The sculptors' village revealed how stone carving is still passed from generation to generation. The trip showed us that heritage is a living tradition, and it taught us to respect and protect our monuments.

Question 28

28. a) Write a formal Letter of Request to the Librarian of your college seeking permission to extend library working hours during the semester examination period. Explain the academic benefits, the expected student participation, and the arrangements required to ensure proper use of library facilities. (13 Marks)

Answer (28.a)

T. Harshavardhan, I Year B.E. CSE,
St. Joseph's College of Engineering, Chennai - 119
Date: 07 November 2026

Dr. M. Kavitha, The Librarian,
St. Joseph's College of Engineering,
OMR, Chennai - 119

Subject: Request for extension of library working hours during the semester examination period

Respected Madam/Sir,

I am writing on behalf of the students of B.E. CSE to request permission to extend the library working hours during the upcoming semester examinations.

During examinations, students need a quiet environment with access to textbooks, reference books, previous question papers and journals. The current closing time limits our preparation, particularly for students who live in hostels or travel long distances. Extended hours until 8:00 p.m. would let us revise systematically, clear doubts through reference books and improve our academic performance.

Nearly 150 students have expressed interest in using the library during the extended hours. To ensure proper use, we suggest that:

- Entry be allowed only on presentation of the college ID card.
- A staff member and a student volunteer be present to maintain discipline.
- Silence be maintained and library property be handled carefully.
- The extension apply only during the examination period.

We assure you of our full cooperation in following all library rules. We would be grateful if you would kindly grant this request. Thank you for your consideration.

Yours faithfully,
T. Harshavardhan

28. b) Draft a formal Letter of Request to the Principal of your college seeking permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution. Mention the objectives of participation, the expected learning outcomes, and the arrangements required. (13 Marks)

Answer (28.b)

T. Harshavardhan, I Year B.E. CSE,
St. Joseph's College of Engineering, Chennai - 119
Date: 07 November 2026

Dr. S. Ramanathan, The Principal,
St. Joseph's College of Engineering,
OMR, Chennai - 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. CSE student of your college. I came to know that IIT Madras is organising a two-day technical workshop on Artificial Intelligence and Emerging Technologies on 20-21 December 2026 at Chennai. I request your kind permission to attend this workshop.

My aim is to gain practical exposure to current AI tools, machine learning concepts and emerging technologies, and to interact with industry experts and researchers.

The workshop will help me build hands-on skills that support my coursework and mini-projects, understand real-world applications of AI, improve my employability, and share the knowledge with my classmates through a short presentation after returning.

I kindly request the following arrangements:

- Permission for on-duty leave for the workshop days.
- A bonafide certificate / recommendation letter for registration.
- Permission to travel to the venue; I will bear the registration fee and travel cost.

I assure you that I will make up for any missed classes and abide by the rules of the host institution. I would be grateful if you would kindly grant me permission.

Yours obediently,
T. Harshavardhan

Question 29

29. a) Compose an analytical essay (250-300 words) comparing the advantages and disadvantages of Artificial Intelligence-Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. (13 Marks)

Answer (29.a)

Assessment shapes how engineering students learn, and the rise of Artificial Intelligence has created a debate between AI-based and traditional teacher-based evaluation.

AI assessment offers clear advantages in accuracy and speed. Automated systems can grade objective tests, programming tasks and numerical problems consistently, without fatigue or arithmetic errors. Feedback arrives within seconds, so students can correct mistakes while the topic is fresh, whereas teachers may take days or weeks to return scripts. AI is also free from personal bias such as mood, handwriting preference or favouritism, which supports fairness when the algorithm is well designed.

However, AI has serious limitations. It struggles with creativity, original reasoning and open-ended design answers, where marks must be awarded for partial understanding. Algorithms trained on biased data can themselves be unfair, and they may penalise unusual but valid solutions. Over-reliance on technology also raises concerns about data privacy and technical failure.

Teacher-based assessment excels in human judgement. Teachers understand context, recognise effort, interpret unconventional approaches and give motivational, personalised feedback. Yet it is slower, prone to subjective bias, and inconsistent when many evaluators are involved. Heavy workloads can also reduce the quality of comments, and large classes make truly individual attention difficult.

In terms of accuracy and feedback time, AI clearly leads; in judging depth, creativity and fairness to individual circumstances, teachers remain superior. The most effective approach is a hybrid model in which AI handles routine, objective evaluation while teachers assess projects, reports and viva performance. Such a combination preserves efficiency without losing the human touch essential to engineering education, and it builds student trust in the assessment process.

29. b) Write a compare-and-contrast essay of 250-300 words analysing the advantages and disadvantages of Living in Urban Areas versus Rural Areas. Discuss employment opportunities, cost of living, access to education and healthcare, and quality of life. (13 Marks)

Answer (29.b)

Where a person lives strongly influences career, health and happiness. Urban and rural areas offer very different advantages, and choosing between them involves real trade-offs.

Employment is the most visible difference. Cities provide a wide range of jobs in industry, information technology, services and research, along with better salaries and career growth. Rural areas depend mainly on agriculture and small trades, so opportunities are limited and incomes are often seasonal, which forces many young people to migrate.

Cost of living, however, favours villages. Rent, food, transport and daily expenses are much higher in cities, and many families struggle with small, crowded homes. Rural residents enjoy cheaper housing, fresh local produce, lower expenses and often more savings at the end of the month.

Access to education and healthcare is again stronger in urban areas. Cities have reputed schools, colleges, coaching centres, and well-equipped hospitals with specialist doctors. Villagers often travel long distances for higher education or emergency treatment, although government schemes and telemedicine are gradually reducing the gap.

Quality of life presents a mixed picture. Urban life offers convenience, entertainment and better infrastructure, but it also brings pollution, traffic, noise and stress. Rural life provides clean air, open space, close community bonds and a calmer pace, but it may lack modern amenities and reliable internet.

In conclusion, cities are better for career prospects, education and healthcare, while villages offer affordability, peace and a healthier environment. The ideal choice depends on individual priorities, and balanced regional development, with better jobs and facilities in rural areas, can narrow the difference between the two.

Question 30

30. a) Draft a personal email to your school friend sharing your experiences during the first semester of your engineering course. Describe your new friendships, classroom activities, adjustment to college life, and expectations for the upcoming semester examinations. (13 Marks)

Answer (30.a)

To: karthik.s2008@gmail.com

Subject: My first semester at engineering college: all the news!

Hi Karthik,

How have you been? I'm sorry for the delay in writing. The first semester has been so busy that time just flew by, but I wanted to share my experiences with you.

The biggest joy has been the new friendships. I now have a close group of friends from different districts, and we study together in the library, share notes and cheer each other up before tests. Our classmates come from varied backgrounds, which has taught me a lot.

Classes are quite different from school. We have theory subjects like Engineering Mathematics and Physics, and lab sessions where we handle real equipment, write programs and prepare reports. I especially enjoy the practical classes, as concepts suddenly make sense when we see them working. Our faculty also encourage presentations and group projects, which have improved my confidence.

Adjusting to college life was not easy at first. Managing time, assignments and attendance without constant reminders from teachers was a challenge, but I soon learnt to plan my week and balance studies with sports and clubs.

The semester examinations begin soon. I am revising regularly, solving previous question papers and clearing my doubts with seniors. I am a little nervous, but I hope to score well.

Do write and tell me about your college too. Let's meet during the vacation!

Yours always,
Harshavardhan

30. b) Draft a personal email to your school friend describing your experience attending the orientation programme for first-year engineering students at your college. Highlight the activities conducted, the introduction to academic facilities and college regulations, and how the programme helped you understand college life and prepare for your academic journey. (13 Marks)

Answer (30.b)

To: arun.prakash08@gmail.com

Subject: My first-year orientation at college: you'd have loved it!

Hi Arun,

Hope you're doing well! It's been ages since we last talked, so I thought I'd write and tell you about my orientation programme at college.

It was held over three days for all first-year engineering students. On the first day, the Principal welcomed us and spoke about the college's vision and the opportunities ahead. Senior students and faculty then led icebreaker games, which helped me make friends from different departments quickly. We also had a motivational talk on goal-setting and time management.

The next day, we were taken on a tour of the facilities: laboratories, the central library, workshops, the Wi-Fi campus and sports grounds. Our department head explained the syllabus, the credit system, internal assessments and the attendance rules, along with the college's code of conduct. Honestly, hearing all the regulations clearly made me feel more prepared and less nervous.

What I liked most was the interaction with seniors, who shared tips about studying, projects and clubs. By the end, college life felt much less intimidating, and I understood the discipline and effort needed for the academic journey ahead.

Tell me about your college too! Write back soon, and let's plan to meet during the holidays.

Take care,
Harshavardhan

CEFR English Level Assessment

Preliminary English Placement Report

A2–B1z

Elementary to lower-intermediate English

1. Assessment results

Assessment	Result
Grammar	3/5
Reading comprehension	3/3
Self-introduction	Clear ideas; spelling and grammar need improvement
Opinion writing	Logical ideas with developing sentence complexity

2. Writing evaluation

- Grammar: Practise verb forms, prepositions, and sentence structure.
- Vocabulary: Good use of words such as opportunities, efficiently, and security.
- Coherence: Ideas are arranged logically.
- Complexity: You can connect ideas using however and if.
- Spelling and punctuation: Review spelling, capitalisation, commas, and full stops.

3. Corrected writing sample

I think AI will create more opportunities than problems in the future. It can help people learn new skills, solve difficult problems, and complete tasks more efficiently. However, AI may also replace some jobs and create security risks. If we use AI wisely, it can make our lives better.

4. Recommendations

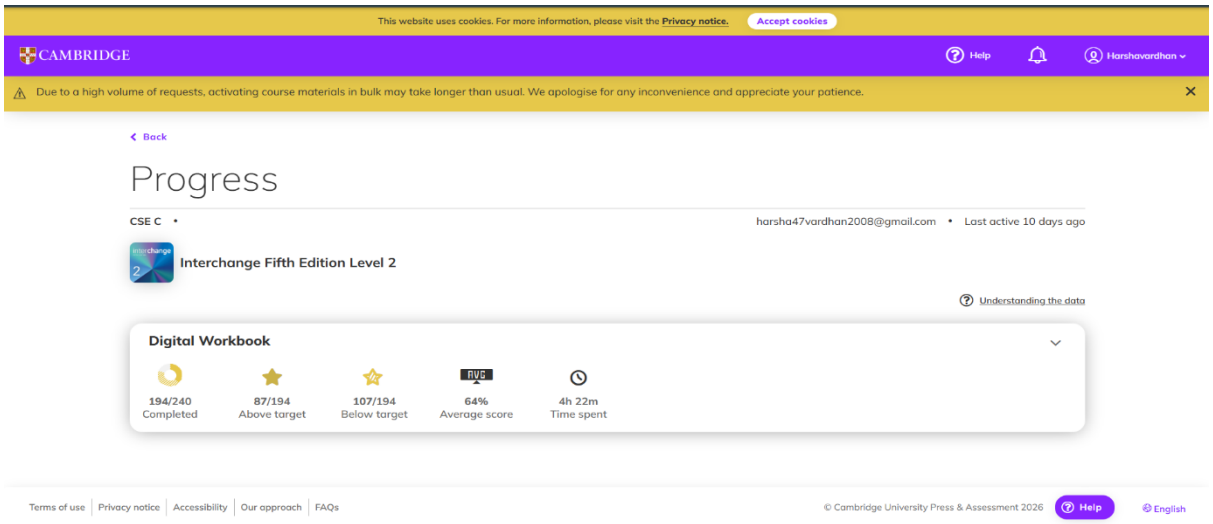
1. Practise English grammar for 15 minutes daily.
2. Write a short paragraph every day.
3. Learn five new English words daily.
4. Practise speaking about familiar topics for one minute.
5. Use linking words such as although, because, and therefore.

5. Final result

Estimated level: A2 moving towards B1.

This is an informal estimate, not an official CEFR certification. Listening and live speaking were not assessed, so further testing may change the result.

CAMBRIDGE PROGRESS



LinkedIn Profile

Contact

7/59 Eswaran Koil street
Alandur
Chennai-16
9345356435 (Mobile)
harsha47vardhan2008@gmail.com

www.linkedin.com/in/harsha-vardhan08 (LinkedIn)

Top Skills

C (Programming Language)
HTML
JavaScript

Harsha Vardhan

Student at St. Joseph’s College Of Engineering B.E. CSE Student | C Learner | HTML & JavaScript Basics | Cybersecurity & AI/ML Enthusiast
Chennai, Tamil Nadu, India

Summary

I am a first-year B.E. Computer Science and Engineering student with a growing interest in Cybersecurity and Artificial Intelligence/ Machine Learning. I have basic knowledge of HTML and JavaScript, and I am currently learning C programming and strengthening my programming fundamentals.I enjoy exploring new technologies, learning through hands-on practice, and taking part in technical events, hackathons, and projects. While continuously improving my technical, problem-solving, and programming skills.

Education

Amrita Vidyalayam nesapakkam
12th qualified, Bio maths · (April 2024 - March 2026)

St. Joseph's College Of Engineering
Computer Science · (August 2026 - May 2030)

GitHub Profile

Computer Science Engineering Student | Technology Enthusiast

ABOUT ME

- BE Computer Science Engineering student.
- Learning C Programming, HTML and JavaScript.
- Interested in Cybersecurity.
- Exploring Artificial Intelligence and Machine Learning.
- Interested in building innovative technology solutions.
- Always curious to learn new things.

TECHNICAL SKILLS

C Programming	**HTML**	**JavaScript**	**Git & GitHub**

Areas of interest: Cybersecurity | Artificial Intelligence | Machine Learning

PROJECTS

PROJECT	DESCRIPTION
Mini Calculator	A simple calculator program developed using C.
AI-Powered Campus Operating System	A smart campus project concept designed to bring student, faculty and campus services together.
More Projects	More projects and practical learning work will be added as they are completed.

GITHUB PROFILE

GitHub: github.com/Harsha-08-vardhan

GitHub statistics and most-used language cards can be displayed dynamically on the online README.

MY GOALS

- Improve my programming skills.
- Explore Cybersecurity and AI/ML.
- Build real-world projects.
- Participate in hackathons.
- Become a skilled technology professional.

CONNECT WITH ME

GITHUB	github.com/Harsha-08-vardhan
LINKEDIN	linkedin.com/in/harsha-vardhan08/

Thank you for visiting my profile!